

Central Clearing

12.1 DIRECT CLEARING

Central clearing replaces the $n(n - 1)/2$ possible bilateral clearing relationships that are possible between n market participants with only n bilateral relationships, each between a market participant and a central counterparty (CCP¹). This simplification of relationships goes beyond the mere operational efficiency which comes with the netting possibilities shown in Figure 4.3 on page 43. Each bilateral relationship requires the setting up and maintenance of payment instructions, as well as credit risk management.

Central clearing is a risk-mitigation tool that is most effective for trades that have a long lifespan. This is because a longer lifespan means that a trade can move further from its inception value and therefore create counterparty performance risk. However, at large volumes even short-dated trades, such as the regular T+2 settlement of a repo transaction, can create substantial balance sheet exposures for financial firms. Consequently, central clearing is now pervasive across the financial industry.

Mechanically, trade negotiation between two counterparties is done as normal, but immediately after the trade has been concluded it is 'given up' to the CCP as shown in Figure 12.2. For trades conducted on an exchange, the give-up to the clearing house of the exchange is generally automatic. In the European Union, new regulations demand that clients can specify that a trade is cleared at a CCP of their choosing (so-called 'client-directed clearing'). In practice, derivatives exchanges remain the exclusive clearers of their contracts and have obtained temporary regulatory exceptions to remain in this position.

The advantages of central clearing among frequent trades are sufficiently obvious to have led to a long history of this technique. The Osaka Rice Exchange established in 1697 used central clearing and margining in a similar fashion to modern CCPs.

At the same time, the presence of a central counterparty creates a cost and introduces new types of risk. The cost is related to the setting up and operation of the CCP. Given that the CCP is counterparty to every trade cleared there, a failure of the CCP to perform under a contract would have catastrophic consequences for financial stability. Because the CCP itself does not conduct trades, and therefore does not endogenously

¹Central counterparties are treated by regulators as a form of market infrastructure and this is the keyword to look for when searching for regulatory details.

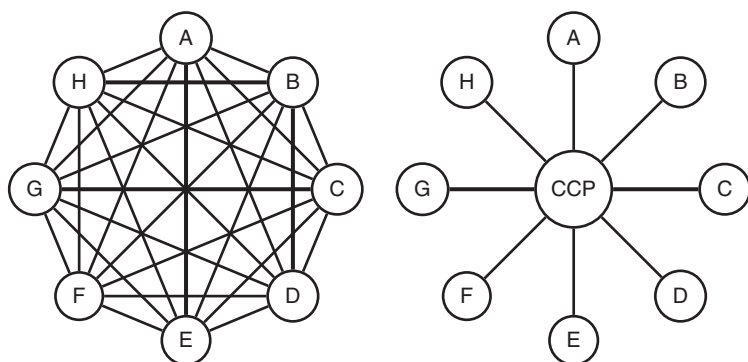


FIGURE 12.1 Clearing relationships with bilateral and central clearing. The 28 possible bilateral relationships between 8 counterparties collapse into just 8 when a CCP is used.

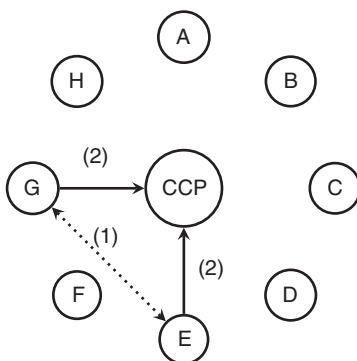


FIGURE 12.2 Trade execution with central clearing. After the trade is concluded (1), it is immediately given up to the CCP (2).

generate any risk of defaulting on a trade-related obligation, the risk for the CCP is that one of its counterparties fails to perform and so the CCP does not have the resources to pay out on the other side of the affected trades. Capitalising a CCP to such a degree that any conceivable risk could be covered with the CCP's own capital would require an economically prohibitive setup cost. This means that measures to reduce the risk exposure of CCPs are critical design features of such arrangements. As will be explained below, some of these mitigation measures create new types of risk for the direct counterparties of the CCP. Such a counterparty is known as a clearing member.

The first step in reducing CCP risk is the frequent exchange of cash margin, known as margining. This means that every trade is revalued on a daily basis using a universally² agreed pricing methodology. Such margin payments related to revaluation of existing positions are known as variation margin (VM). Margining in practice

²The word 'universally' here refers to all direct and indirect users of the CCP.

constrains the type of products that can be cleared centrally to those that on one hand can be priced reliably every day, and on the other hand have sufficiently low volatility to make margin payments practical. Potentially highly volatile derivatives, such as credit default swaps, are therefore difficult to clear, as are exotic derivatives. In essence, variation margining transforms counterparty risk into liquidity risk.

Variation margin is only a partial protection against the implications of market price movements under default risk. It takes some time for a clearing house to deal with a default of a clearing member. During that time, market prices are likely to move further against the position of the defaulting member but that clearing member will no longer be in a position to contribute additional variation margin. Clearing houses therefore charge an additional margin, called initial margin³ (IM) for new positions. Initial margin is therefore calibrated relative to the expected market movement during an assumed time period corresponding to the time it takes to unwind a defaulting member's positions. In the simplest case, one would set the margin to $\rho\sqrt{t}$ where ρ is the one-day expected variance of a given position and t is the number of days assumed necessary to close out a defaulting member's positions. In practice, CCPs set initial margins in a more complex way to reflect partially offsetting risks. For instance, a clearing house would not simply add initial margins for a paid 11 year swap, received 12 year swap position in the same currency given that the 11 and 12 year swap rates are highly correlated and the positions therefore partially offsetting. CCPs to some extent compete on the efficiency of their margining systems. A CCP that can more realistically reflect the risks of a given portfolio will be more attractive to users than one that charges apparently unnecessary margin against less risky positions. At the same time, stable correlations are the precondition for such risk offsets. Regulators therefore restrict cross-position netting at CCPs to prevent artificially low margin requirements.

While variation margin must generally be posted, and is paid out, in cash, clearing houses accept non-cash collateral for initial margin. As the return on margin is due to the clearing member posting it, this arrangement ensures that the more permanent initial margin is not an undue drag on the income of the posting member. However, the interaction between variation and initial margin is more complex. Large market movements create by themselves a need for rapid variation margin payments and receipts as the CCP revalues existing positions. At the same time, such events change the estimates of market volatility and can therefore lead to increased initial margin.

Margining means that any accrued profit or loss on a transaction (which for the CCP, like any other counterparty, means a claim or obligation to pay, respectively) is frequently settled through cash payments. This resets the exposure value of the trades, and therefore the exposure of the CCP to zero. The reason that CCP margins are paid in cash is that this makes the payments immediately final⁴. Frequent margining also means that the ability to pay of any clearing member is tested frequently. A gradual deterioration of the credit quality of a clearing member will therefore eventually lead to a default on a margin call at a time when there may still be assets to mitigate losses.

³In bilateral clearing with margining, this margin is known as the independent amount (IA).

⁴Margining is also used in bilateral (non-cleared) trades, and there margin can also take the form of other collateral. In essence, such arrangements reflect the risk management preferences and asset ownership preferences of the two parties involved.

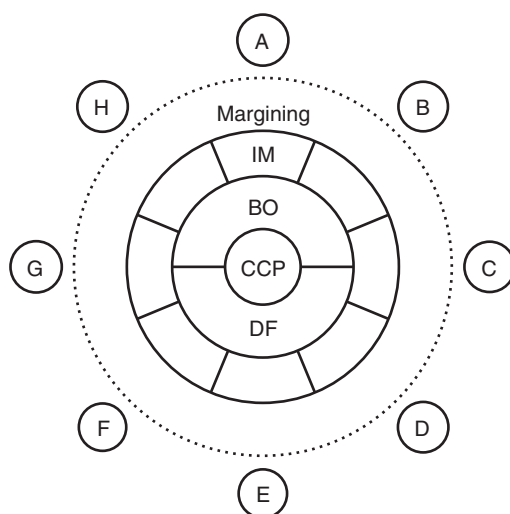


FIGURE 12.3 Standard protection mechanisms for a central counterparty. After margining, the CCP is protected by the initial margin (IM) posted by clearing members, and furthermore by the default fund (DF) and the bidding obligations (BO). Only after all these buffers are exhausted would the CCP itself be threatened by default.

Clearing members can default. Essentially there are two fundamentally different types of such defaults. The first is an external default where the member is declared insolvent by a court or declares insolvency. In this event, there is no uncertainty about whether or not that member can fulfill margin obligations to the CCP because it will simply be legally unable to do so. The second is a failure to make variation margin payments to the CCP as and when required. In such internal defaults, it is essentially up to the CCP to enforce its claims on the member (potentially resulting in an external default) or to postpone such claims. CCPs generally have decision-making structures in place to balance the risks of enforcing an external default or allowing obligations to build up while margin claims remain unmet.

Once a clearing member does default, the CCP is faced with potential losses from unfulfilled payment obligations. The layer of protection after variation margin is the initial margin posted by the clearing member to the CCP. Once a clearing member defaults, the initial margin assets can be realised to pay out variation margin to other clearing members until the positions of the defaulting members are closed out⁵. While a clearing member is in default but the CCP has not yet entered default proceedings, the CCP can pay out variation margin by borrowing against IM assets posted by the member that is failing to pay VM.

The sizing of initial margin is dependent on the prospect of closing out a defaulted clearing member's positions in a set amount of time, and at a maximum loss. This requires that there is a market for such positions at the time the default occurs. To ensure

⁵Once these positions are closed, they cease to generate variation margin calls.

the existence of such a market, CCPs impose a bidding obligation on clearing members at closing auctions of batches of positions held by a defaulting member. After a default, the CCP will typically batch positions by maturity and currency and then require the remaining clearing members to bid for these positions. Because the market will have a general idea of the net positions of the defaulted member and therefore move against these positions, the amounts bid will usually be negative, implying that the bidding member asks to be paid to take on the position of the defaulted member. The initial margin posted by the defaulted members should go some way towards covering this cost. Should that be insufficient, the CCP can dip into the default fund, a segregated pool of assets contributed by all clearing members, to cover the amount not already covered by the initial margin posted by the defaulted member. Some CCPs also mandate clearing members to assist with personnel for the closing of positions, meaning that experienced trading staff is posted to the CCP for a short time to execute position transfers.

The mutual nature of the default fund creates a natural incentive for clearing members to post low bids in the closing auctions. A low bid ensures a low likelihood of incurring losses on the future evolution of the positions acquired in the auction while the lower valuation is absorbed by the mutual default fund. This incentive is counteracted by CCPs through an obligation to replenish the default fund after a closing auction. The less a clearing member bids for positions, the more that member has to contribute to the replenishment of the default fund.

Central counterparties need to optimise the sizing of the various protection components with a view to achieving an acceptable level of residual risk with a minimum level of deployed capital. The outcome of this optimisation is dependent on market volatility as well as the type, volume and distribution of business going through the CCP at any one time. CCPs typically run daily stress tests simulating the default of their largest clearing member and adjust margin requirements in response to changes in market volatility.

Bidding obligation and default fund contributions create contingent liabilities for clearing members that are crystallised by the default of another member. In addition, the requirement to post variation margin in cash, and at short notice, can create liquidity stresses. These risks are fundamentally different from usual bilateral trading relationships because it is not clear which clearing member default would trigger which type of close-up auction, and the exposure of the CCP to any given clearing member is only known to that member. The potentially required top-up amount for the default fund is therefore nearly impossible to establish for a clearing member.

As a result of these complex risks, clearing membership creates complex capital requirements for banks. At the same time, many financial and non-financial institutions are unable to become clearing members because they are unable to manage such risks, or do not have, as would for example be the case for pension funds, any capital that could be committed to clearing member obligations. For this reason, central clearing is often accessed through intermediaries. The complexity of risk management do however appear to lead to a rationing of such services rather than to supply that reacts elastically to pricing.

12.2 INDIRECT CLEARING

When a market participant is unable to become a clearing member of a CCP, central clearing can be done indirectly through the services of a clearing member. The clearing member is then said to act as clearing agent for the non-member. This choice of words creates some unfortunate room for confusion because there are at this time two models for the provision of clearing services known as agency clearing and principal clearing. These two models differ in the relationship between the CCP and the non-clearing member and the choice between them is driven more by history than economics. At the time of writing, the US market is dominated by agency clearing while the European and Japanese market is dominated by principal clearing.

An essential new problem introduced by indirect clearing is the separation of obligations of the clearing member and the client. In principal clearing, all obligations towards the CCP are with the clearing member while they are split between clearing member and non-clearing member in the agency model. These differences concern the risks faced by the CCP.

At the same time, non-clearing members face the risk that their clearing agent defaults while holding their assets. In such a situation, assets may be encumbered with the clearing agent while the market risk is in effect still borne by the non-clearing members. An essential element of CCP default management is therefore the transfer of positions from a defaulting clearing agent to non-defaulting agents. The problem here is that clearing agency is a bilateral relationship that is in principle outside the purview of the CCP because it involves the full gamut of bilateral credit risk and lending arrangements possible between a liquidity provider and a client. An essential safeguard in such situation is the segregation of margin assets between the clearing agent and the non-clearing clients.

12.2.1 Agency clearing

In the agency clearing model, the CCP has contractual relationships with non-clearing users but the essential obligations of clearing are delegated to clearing members. This means that the clearing member contracted by a non-clearing user is responsible for posting margin and bidding in close-out auctions. However, the CCP has full visibility on the total positions of every end-user. The services provided by the clearing member acting as agent consist of ensuring that variation margin calls are met and it is between that clearing member and its client to arrange for either delivery of sufficient cash, or the transformation of non-cash collateral into cash through repurchase transactions.

The fundamental benefit of agency clearing is that the CCP has full information about the individual positions held by all beneficial owners of all trades cleared on the CCP. This makes it possible to conduct stress testing on the basis of ultimate ownership of positions rather than on indirect information which may be skewed when non-clearing members use multiple clearing agents.

The downside of agency clearing is that substantial obligations remain with non-clearing users but the CCP has limited information about these entities. In particular, the CCP will usually not be able to judge the ability of a non-clearing member to post sufficient collateral to its clearing agents in case of large market movements.

While this type of risk is in principle the responsibility of the clearing members to manage, the survival of the CCP potentially depends on this type of assessment.

12.2.2 Principal clearing

In the principal clearing model, the CCP has only clearing members and no contractual relationships with non-clearing users. Clearing members act as agents that stand completely between the CCP and the end client. In essence, a clearing member acts as if its own positions and those of its non-member clients form a single block with all attendant obligations. As far as the CCP is concerned, non-clearing members simply do not exist.

In the principal clearing model, CCPs typically offer convenience accounts to clearing members that allow detailed separation between the positions maintained by the clearing member on behalf of individual clients, but there is no pretence that such positions can be aggregated across multiple clearing agents used by the same end-user. In other words, the CCP cannot identify, much less quantify, positions maintained by one non-clearing user who splits clearing business across multiple clearing members. As in the agency clearing model, this exposes the CCP to risks related to defaults of large non-clearing users, or, equivalently, the correlated default of multiple non-clearing users with similar positions.

12.2.3 Hybrid clearing models

Both agency and principal clearing are characterised by a clear separation between clearing members and non-clearing members of a CCP. They differ in whether or not non-clearing members have an explicit contractual relationship with the CCP.

With mandatory central clearing, essentially shortcomings of central clearing have become more apparent. To be a clearing member, a market participant has to have ready access to central bank cash and be able to transfer such cash in a timely manner to and from the CCP. The member must also have sufficient equity to commit to the default fund and have the risk-management expertise to assist in the close-out process of defaulted members. This can entail the management of residual positions which may have contractual maturities extending into decades. While these features are generally in place for financial trading firms, they are less prevalent among many entities that have been mandated in recent years to clear positions, e.g., pension funds or government entities. It is difficult to envision a tax-funded entity managing the 30 year swap book of a defaulted investment bank.

The standard solution, which is to have non-clearing members contract out their obligations to clearing members under the principal or agent clearing model, is limited by the availability of balance sheet at these clearing members. Total market risk is not reduced by redistributing contingent risks from one type of market participant to another. Indeed, given the specific requirements of clearing members, concentrating the liquidity risks of the entire market on a limited set of clearing members is likely to increase, not decrease, financial stability risks. Clearing members tend to prudently ration their provision of clearing services which creates a bottleneck for the clearing of positions. In particular, entities that do not normally hold cash and do not have access to money

market or central bank borrowing have to rely on their ability to access the repo market in order to mobilise cash for variation margin payments. The general market risk of non-performance of such entities is therefore not eliminated but only transformed into the risk of their inability to access the repo market in stressed scenarios.

CCPs have responded to this bottleneck by establishing clearing models that separate different obligations of clearing members, for instance Eurex Clearing's ISA Direct model [21]. Under such hybrid models, entities manage the initial and variation margin process themselves but contract the default management obligation to financial services firms. Under current capital requirement and liquidity coverage rules, relieving clearing members from the collateral transformation obligations of the principal clearing model makes a substantial economic difference. However, given the jump risk nature of the default management process, it is not immediately clear that this release of regulatory capital corresponds to an actual decrease in risk.

12.3 CONTRACT VALUE ADJUSTMENTS (XVA)

Central clearing is now the default clearing mechanism for most liquid derivatives contracts, and for many spot transactions including repos. Therefore the prices in the central clearing market are probably the best reference for market prices.

This then raises the question of how to value derivatives that are not cleared centrally. Theoretically, clearing has no impact on valuation because accounting standards specify (here in the phrasing of AG71 of IFRS 39 [48]): 'Fair value is defined in terms of a price agreed by a willing buyer and a willing seller in an arm's length transaction.' In practice, however, different clearing models do imply different costs in terms of capital or margin. More significantly, the value of any one specific transaction can depend on all other transactions that are pending with the same counterparty. This is because each new transaction will change the future expected exposure versus the counterparty and so, if applicable, the expected amount of margin or collateral that needs to be posted in the future.

Financial institutions therefore need to apply valuation adjustments, collectively known as xVA, where the letter x stands for other letters in abbreviations like CVA, FVA, DVA, etc., that will be explained in more detail below.

12.3.1 Credit Value Adjustment

When a transaction is not cleared or margined, any mark-to-market revaluation exposes the party with a positive mark-to-market to default risk of the other counterparty.

Simply speaking, the transaction may have a positive market value, but the risk is that the other counterparty fails to perform as promised.

Traditionally, banks trading derivatives projected future mark-to-market exposures against their counterparties and charged expected exposures against the credit lines approved for these counterparties. With risk-based capital requirements, these credit exposures incur capital charges, and the cost of this capital needs to be assigned to the transaction. This cost is called the credit value adjustment, or CVA.

Structurally, the CVA depends not only on the transaction (as accounting standards would seem to imply), but also on the credit risk of the counterparty, the capital cost of the bank as the other counterparty, and all other transactions between these two counterparties. The latter link arises because a new transaction that reduces future expected exposure between the two counterparties reduces the capital charge and so should carry a negative CVA. While economically correct, this notion contradicts the fundamental idea of accounting fair value because the new transaction has a capital impact that is unique to the two contracting parties. The notion of an 'arm's length transaction' is therefore not applicable in this context. In addition, the notion of a generic 'willing buyer and willing seller' neglects the specific credit risk and capital cost essential to the CVA calculation.

Swap dealers tend to hedge large-scale counterparty risks through the use of credit default swaps. This means that CVA for large counterparties is proxied fairly well through CDS spreads.

12.3.2 Funding Value Adjustment

When a transaction is margined, CVA is no longer appropriate because the credit risk aspect of the transaction is addressed through margining. Instead, counterparties must concern themselves with the cost of providing margin to each other. While CVA incurs a cost to the counterparty that has a positive mark-to-market (due to the credit risk it faces), margining creates a cost to the counterparty that has a negative mark-to-market because it needs to post margin. Margin posted is remunerated by the receiving party.

Margin is usually treated as a cash equivalent, and is remunerated at a cash rate, while the party posting it needs to raise the margin at its own funding cost, the funding value adjustment (FVA) can be sizeable. For instance, a bank funding itself unsecured at an average rate of 50 bps over the overnight average (SOFR, €STR, SONIA, etc.) would have a running cost of 50 bps for cash posted as margin if that cash is remunerated at the overnight rate.

MVA in practice is more complex because margin can in some instances be posted in multiple currencies. The party posting the margin will choose the currency that offers the most attractive remuneration, taking into account cross-currency basis swap rates. The future remuneration rate of posted margin can therefore be a complex option involving funding spreads in different currencies.

FVA can lead to situations where a single derivatives dealer has multiple positions that offset each other in terms of risk, but differ in the margining provisions. This means that the dealer has no market risk, but funding risk resulting from mismatches in how

margin is received and posted. Such funding costs can be sizeable relative to the potential profit margins in liquid derivatives markets.

12.3.3 Debit Value Adjustment

In some clearing arrangements, collateral requirements depend on the rating of the counterparties. When downgraded below certain trigger levels, collateralisation, or an increase in collateralisation levels, becomes mandatory in such contracts. This means that a counterparty in effect faces higher costs when its own credit quality deteriorates. Counter-intuitively, this means that financial firms may face an economic need to hedge their own credit risk. Debit value adjustments (DVAs) reflect the economic cost of this obligation to the best estimate of the firm applying it, bearing in mind that an actual hedge is virtually impossible.